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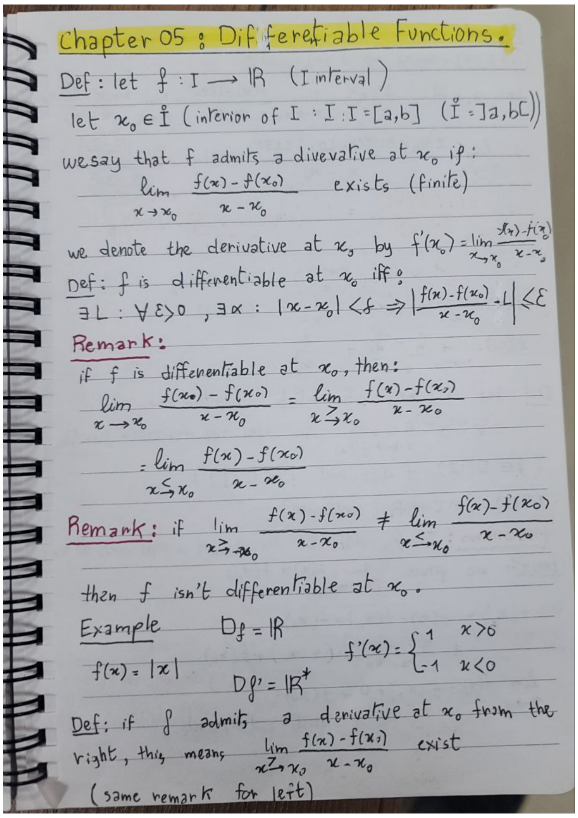
Chapter 5 Differetiable functions

Tags	Courses



Chapter 05 : Differetiable Functions.

Def: let $f : I \longrightarrow \mathbb{R}$ (I interval)

let $x_0 \in \mathring{I}$ (interior of I : I : $I = [a,b]$ ($\mathring{I} =]a,b[$))

we say that f admits a divevative at x_0 if:

$$\lim_{x \to x_0} \frac{f(x) - f(x_0)}{x - x_0} \text{ exists (finite)}$$

we denote the derivative at x_0 by $f'(x_0) = \lim_{x \to x_0} \frac{f(x) - f(x_0)}{x - x_0}$

Def: f is differentiable at x_0 iff:

$\exists L : \forall \varepsilon > 0, \exists \alpha : |x - x_0| < \delta \Rightarrow \left| \frac{f(x) - f(x_0)}{x - x_0} - L \right| \leq \varepsilon$

Remark:

if f is differentiable at x_0, then:

$$\lim_{x \to x_0} \frac{f(x) - f(x_0)}{x - x_0} = \lim_{x \overset{>}{\to} x_0} \frac{f(x) - f(x_0)}{x - x_0}$$

$$= \lim_{x \overset{<}{\to} x_0} \frac{f(x) - f(x_0)}{x - x_0}$$

Remark: if $\lim_{x \overset{>}{\to} x_0} \frac{f(x) - f(x_0)}{x - x_0} \neq \lim_{x \overset{<}{\to} x_0} \frac{f(x) - f(x_0)}{x - x_0}$

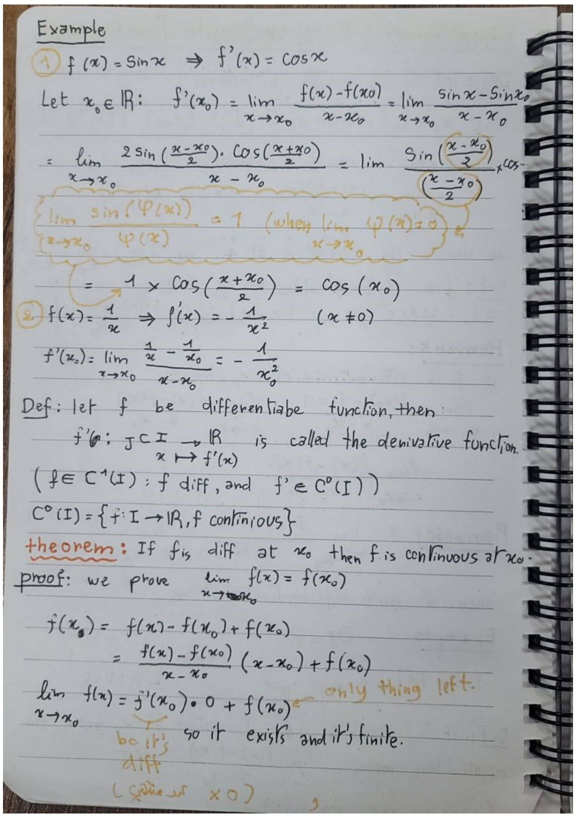
then f isn't differentiable at x_0.

Example $D_f = \mathbb{R}$

$f(x) = |x|$ $\quad D_{f'} = \mathbb{R}^*$ $\quad f'(x) = \begin{cases} 1 & x > 0 \\ -1 & x < 0 \end{cases}$

Def: if f admits a derivative at x_0 from the right, this means $\lim_{x \overset{>}{\to} x_0} \frac{f(x) - f(x_0)}{x - x_0}$ exist

(same remark for left)



Example

1. $f(x) = \sin x \Rightarrow f'(x) = \cos x$

Let $x_0 \in \mathbb{R}$: $f'(x_0) = \lim_{x \to x_0} \frac{f(x) - f(x_0)}{x - x_0} = \lim_{x \to x_0} \frac{\sin x - \sin x_0}{x - x_0}$

$$= \lim_{x \to x_0} \frac{2 \sin\left(\frac{x - x_0}{2}\right) \cdot \cos\left(\frac{x + x_0}{2}\right)}{x - x_0} = \lim \frac{\sin\left(\frac{x - x_0}{2}\right)}{\left(\frac{x - x_0}{2}\right)} \times \cos \cdots$$

$\lim_{x \to x_0} \frac{\sin(\varphi(x))}{\varphi(x)} = 1$ (when $\lim_{x \to x_0} \varphi(x) = 0$)

$$= 1 \times \cos\left(\frac{x + x_0}{2}\right) = \cos(x_0)$$

2. $f(x) = \frac{1}{x} \Rightarrow f'(x) = -\frac{1}{x^2}$ $\quad (x \neq 0)$

$f'(x_0) = \lim_{x \to x_0} \frac{\frac{1}{x} - \frac{1}{x_0}}{x - x_0} = -\frac{1}{x_0^2}$

Def: let f be differentiabe function, then:

$f' : J \subset I \to \mathbb{R}$, $x \mapsto f'(x)$ is called the derivative function.

($f \in C^1(I)$: f diff, and $f' \in C^0(I)$)

$C^0(I) = \{ f : I \to \mathbb{R}, f \text{ continious} \}$

theorem: If f is diff at x_0 then f is continuous at x_0.

proof: we prove $\lim_{x \to x_0} f(x) = f(x_0)$

$f(x) = f(x) - f(x_0) + f(x_0)$

$= \frac{f(x) - f(x_0)}{x - x_0} (x - x_0) + f(x_0)$

$\lim_{x \to x_0} f(x) = f'(x_0) \cdot 0 + f(x_0)$ ← only thing left.

bc it's diff, so it exists and it's finite.

([illegible] × 0)

theorem:

Let $f, g : I \to \mathbb{R}$, let $x_0 \in I$ such that f and g are differentiable. then:

1. αf is diff at x_0 $(\alpha f)'(x_0) = \alpha f'(x_0)$
2. $(f+g)'(x_0) = f'(x_0) + g'(x_0)$
3. $(fg)'(x_0) = f'(x_0)g(x_0) + f(x_0) \cdot g'(x_0)$
4. $\left(\frac{f}{g}\right)'(x_0) = \dfrac{f'(x_0)g(x_0) - f(x_0)g'(x_0)}{g^2(x_0)}$

proof

for 3: $(fg)'(x_0) = \lim\limits_{x \to x_0} \dfrac{(fg)(x) - (fg)(x_0)}{x - x_0}$

using function properties

$$= \lim_{x \to x_0} \frac{f(x) \cdot g(x) - f(x_0)\, g(x_0)}{x - x_0}$$

$$= \lim_{x \to x_0} \frac{f(x)g(x) \overbrace{- f(x_0)g(x) + f(x_0)g(x)} - f(x_0)g(x_0)}{x - x_0}$$

$$= \lim_{x \to x_0} \left(\frac{f(x) - f(x_0)}{x - x_0}\right) \cdot g(x) + f(x_0) \cdot \frac{g(x) - g(x_0)}{x - x_0}$$

$$= f'(x_0)\, g(x_0) + f(x_0) \cdot g'(x)$$

for 4: $\left(\frac{f}{g}\right)'(x_0) = \lim\limits_{x \to x_0} \dfrac{\frac{f(x)}{g(x)} - \frac{f(x_0)}{g(x_0)}}{x - x_0} =$

$$= \lim_{x \to x_0} \frac{f(x)g(x_0) - g(x)f(x_0)}{(x - x_0)\, g(x)\, g(x_0)} = \lim_{x \to x_0} \frac{f(x)g(x_0) - f(x_0)g(x_0) + g(x_0)f(x_0) - g(x)f(x_0)}{(x - x_0)\, g(x)\, g(x_0)}$$

$$= \lim_{x \to x_0} \frac{f(x) - f(x_0)}{(x - x_0)} \cdot \frac{g(x_0)}{g(x) \cdot g(x_0)} - \frac{g(x) - g(x_0) \cdot f(x_0)}{(x - x_0)\, g(x)\, g(x_0)}$$

$$= \frac{f'(x_0)\, g(x_0) - g'(x_0)\, f(x_0)}{g^2(x_0)}$$

theorem:

- x_0 is a local minimum
  $\exists \alpha > 0,\ f(x_0) \leqslant f(x), \forall x \in]x_0-\delta, x_0+\delta[$

- if f is diff at x_0, and if f admits an extremum
  then $f'(x_0) = 0$

proof:

$$f'(x_0) = \lim_{x \to x_0} \frac{f(x) - f(x_0)}{x - x_0}$$

we suppose $f(x_0)$ is the minimum $\left(\begin{matrix} f(x_0) \leq f(x) \\ x \in V(x_0) \end{matrix} \right)$

$$\lim_{x \overset{>}{\to} x_0} \frac{f(x)-f(x_0)}{x-x_0} \geqslant 0 \Rightarrow (f'(x_0) \geqslant 0)$$

$$\Rightarrow f'(x_0) = 0$$

$$\lim_{x \overset{<}{\to} x_0} \frac{f(x)-f(x_0)}{x-x_0} \leqslant 0 \Rightarrow (f'(x_0) \leqslant 0)$$

theorem:

Let $f: I \to \mathbb{R}$ and $g: J \to \mathbb{R}$ be two diff functions at x_0 (resp $g(f(x_0))$) such that $f(I) \subset J$. then:

$$(g \circ f)'(x_0) = g'(f(x_0)) \cdot f'(x_0)$$

proof:

We define

$$h(y) = \begin{cases} \dfrac{g(y) - g(f(x_0))}{y - f(x_0)} & y \neq f(x_0) \\ g'(f(x_0)) & y = f(x_0) \end{cases}$$

h is continuous at $f(x_0)$ because $\lim_{y \to f(x_0)} h(y) = g'(f(x_0))$ (g is diff at $f(x_0)$)

one has $(g \circ f)'(x_0) = g(f(x_0))$

by definition : $(g \circ f) = \lim_{x \to x_0} \dfrac{(g \circ f)(x) - (g \circ f)(x_0)}{x - x_0}$

$$= \lim_{x \to x_0} \frac{g(f(x)) - g(f(x_0))}{x - x_0} = \lim \frac{g(f(x)) - g(f(x_0))}{f(x) - f(x_0)} \times \frac{f(x) - f(x_0)}{x - x_0}$$

$$= g'(f(x_0)) \times f'(x_0)$$

theorem (Rolle (Roll)):

Let $f: [a,b] \to \mathbb{R}$ be st:

1. f continuous on $[a,b]$
2. f differentiable on $]a,b[$
3. $f(a) = f(b) \neq 0$ $f(a) = f(b) = 0$ if $f(x) = g(x) - g($

then there exists at least $x_0 \in]a,b[$; $f'(x_0) = 0$

Proof: As f attains a Maximum and minimum so $\exists x_0 \in]a,b[$, $f'(x_0) = 0$

Example of application:

$g(x_0) = 0$
$\sin x_0 = x_0$

- solve $\sin x = x$ in $[-1,1]$
  we consider new function $g(x) = \sin x - x$ $\begin{cases} g(-1) = -\sin 1 + \\ g(1) = \sin -1 < \end{cases}$

- $x_0 \in]0,\pi[$: $\cos'(x_0) = x$
  $g(x) = \sin x$ $g(0)$, $g(\pi) = 0$
  using Rolle theorem, $\exists x_0 \in]0,\pi[$, $(\sin x_0)' = \cos x_0 =$

theorem (Mean value theorem)

let $f: [a,b] \to \mathbb{R}$ be st:

1. f continuous on $[a,b]$
2. f diff on $]a,b[$

then there exist at least $x_0 \in]a,b[$: $\dfrac{f(b)-f(a)}{b-a} = f'(x_0)$

proof: consider the line function h which joins $(a, f(a))$ and $(b, f(b))$, defined by: $h(x) = f(a) + \frac{f(b)-f(a)}{b-a}(x-a)$

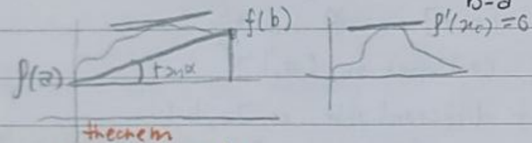
and consider $g(x) = f(x) - h(x)$. Obviously:

g is continuous on $[a,b]$, diff on $]a,b[$

and $g(a) = 0 = g(b)$

so with roll theorem $\Rightarrow \exists x_c \in]a,b[: g(x_c) = 0$

$\Leftrightarrow f'(x_0) - h'(x_0) = 0 \Rightarrow f'(x_0) = \frac{f(b)-f(a)}{b-a}$.



Mean value theorem (extended)

Let f, g be two functions defined on $[a,b]$ (and continuous) and diff on $]a,b[$ st $g'(x) \neq 0$, $\forall x \in]a,b[$ then there exist $x_0 \in]a,b[$ st:

$$\frac{f(b)-f(a)}{g(b)-g(a)} = \frac{f'(x_0)}{g'(x_0)}$$

(same x_0) what's the secret for being the same x_0

$h(x) = (f(x) - f(a))(g(b) - g(a)) - (f(b) - f(a))(g(x) - g(a))$

$h(a) = h(b) = 0 \Rightarrow h'(x_0) = 0 \Rightarrow f'(x_0)(g(b) - g(a)) - g'(x_0)(f(b) - f(a)) = 0$

Hopital's rule:

when we encounter $\lim_{x \to a} \frac{f(x)}{g(x)}$ → $\frac{0}{0}$ or $\frac{\infty}{\infty}$ but we need a condition → $f(a) = g(a) = 0$ and $f'(a), g'(a)$ exist, $g'(a) \neq 0$

ex: $\frac{\sin}{x}$

$$\frac{f(x)-f(a)}{g(x)-g(a)} = \frac{\left(\frac{f(x)-f(a)}{x-a}\right)}{\left(\frac{g(x)-g(a)}{x-a}\right)} = \frac{f'(a)}{g'(a)}$$

strong one: $\lim_{x \to a} f(x) = \lim g(x) = \ell$ [we use $\tilde{f} \begin{cases} f & x \in]a \\ \ell & x = a \end{cases}$

$\frac{f(x) - f(a)}{g(x) - g(a)} = \frac{f'(c)}{g'(c)}$ mean value theorem

[illegible]

↳ example from TD: $f(x) = x^2 \sin\frac{1}{x}$, $g(x) = x$ → $2x \sin\frac{1}{x} - \frac{1}{x} \cos\frac{1}{x}$

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$x \to \infty$ $\frac{\infty}{\infty}$ تغيير متغير

theorem: Let f and g be continuous functions on an interval I containing $x_0 = a$ with $f(a) = g(a) = 0$ if $f'(a), g'(a)$ exist and with $g'(x)$ doesn't vanish (ينعدم) in a neighborhood of a then:

$$\lim_{x \to a} \frac{f(x)}{g(x)} = \lim_{x \to a} \frac{f'(x)}{g'(x)} = \frac{f'(a)}{g'(a)}$$

proof:

one has: $\dfrac{f(x)}{g(x)} = \dfrac{f(x)-f(a)}{g(x)-g(a)} = \dfrac{\left(\frac{f(x)-f(a)}{x-a}\right)}{\left(\frac{g(x)-g(a)}{x-a}\right)}$

so $\lim\limits_{x\to a} \dfrac{f(x)}{g(x)} = \dfrac{f'(a)}{g'(a)}$

$\lim\limits_{x\to a} \dfrac{f(x)}{g(x)} = \dfrac{\lim\limits_{x\to a} f(x)}{\lim\limits_{x\to a} g(x)}$

EXAMPLE:

- $\lim\limits_{x\to 0} \dfrac{1-\cos x}{x^2} = \lim\limits_{x\to 0} \dfrac{\sin x}{2x} = \dfrac{1}{2}$ (hopital Bah Von)

theorem:

suppose that f, and g are continuous on a closed interval $[a,b]$ we suppose f and g are differentiable on $]a,b[$,

- we suppose that $g'(x)$ isn't null near a, and we suppose that $\lim\limits_{x\overset{>}{\to}a} \dfrac{f'(x)}{g'(x)}$ exist then

$\lim\limits_{x\overset{>}{\to}a} \dfrac{f(x)}{g(x)} = \ell$ $\quad (\lim\limits_{x\overset{>}{\to}a} f(x) = \lim\limits_{x\overset{>}{\to}a} g(x) = 0)$

proof

we define $f(a), g(a)$ by the limit of $f(x)$, $g(x)$ when $x\overset{>}{\to}a$ then generalized Mean Value theorem implies $\exists c \in]a,x[$ $a\le b$ st: $\dfrac{f(x)-f(a)}{g(x)-g(a)} = \dfrac{f'(c)}{g'(c)}$

then $\lim\limits_{x\to a} \dfrac{f(x)}{g(x)} = \dfrac{f'(a)}{g'(a)}$.

theorem: Let $f: [a,b] \to \mathbb{R}$ be continuous and differentiable on $]a,b[$. if $f'(x)=0$, $\forall x \in]a,b[$, then f is constant on $[a,b]$.

proof: $f(x)=f(a)$, $\forall x \in]a,b] \Leftrightarrow f$ is constant on $[a,b]$

Let us consider $f:[a,y]$: $y\le b$

f is continuous, and differentiable on $]a,y[$

By MVT, there exist $x_0 \in]a,y[: f(y)-f(a) = (y-a)f'(x_0)$

but $f'(x_0)=0$, which implies $f(y)=f(a)$, $\forall y \in]a,b]$

theorem: Let $f:[a,b] \to \mathbb{R}$ be continuous and diff on $]a,b[$. we have:

1. $f'(x)\ge 0 \Leftrightarrow f$ is non-dec
2. $f'(x)\le 0 \Leftrightarrow f$ is non-inc

proof:

we just proof ①:

$f'(x)\ge 0$, $\forall x \in]a,b[\Rightarrow f$ is non-dec on $[a,b]$

Let $x,y \in [a,b]$, $(x<y)$, then on $[x,y]$, MVT $\Rightarrow$

$\exists x_0 \in]x,y[: \dfrac{f(y)-f(x)}{y-x} = f'(x_0) \ge$

$\Rightarrow f(y)-f(x)\ge 0$

$f(y) \ge f(x)$

Let us now suppose f is non dec on $[a,b]$

$\forall x,y \in [a,b]$, $x\le y \Rightarrow f(x)\le f(y)$ so:

$\dfrac{f(y)-f(x)}{y-x} \ge 0 \Rightarrow \lim\limits_{y\to x} \dfrac{f(x)-f(y)}{y-x} = f'(x) \ge 0$

x is any given on $]a,b[$

(we can prove ② with the same technique)

theorem:

Let $f:[a,b] \to [c,d]$ which is bijective if f is differentiable at x_0, and f^{-1} is diff at $f(x_0)$

then : $f^{-1\prime}(f(x_0)) = \dfrac{1}{f'(x_0)}$ (تركيب)

$(f^{-1} \circ f)(x_0) = x_0 \rightarrow f^{-1\prime}(f(x_0)) \times f'(x_0) = 1$

$f^{-1\prime}(f(x_0)) = \dfrac{1}{f'(x_0)}$

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Higher derivatives

Def: Let f be a function defined on D_f. we say that f admits a derivative of order n, or f is n^{th} diff if $f^{(1)}, \dots, (f^{(n)})$ exist

Example

$f(x) = e^x$, $f^{(n)}(x) = e^x$

$f(x) = \sin x$, $f^{(n)}(x) = \sin\left(x + \frac{n\pi}{2}\right)$

$f(x) = \cos x$, $f^{(n)}(x) = \cos\left(x + \frac{n\pi}{2}\right)$

Def:

We denote by $C^{(n)}(\Omega)$, the set of functions which are n-times differentiable on Ω, with $f^{(n)}$ continuous.

EXAMPLE:

$$f(x) = \begin{cases} x^2 \sin\frac{1}{x} & , x \neq 0 \\ 0 & , x = 0 \end{cases}$$

$$f'(x) = \begin{cases} 2x\sin\frac{1}{x} - \cos\frac{1}{x} & , x \neq 0 \\ \lim\limits_{x\to 0} \dfrac{x^2 \sin\frac{1}{x} - 0}{x - 0} = 0 & , x = 0 \end{cases}$$

$$\lim_{x \to x_0} \frac{f(x) - f(x_0)}{x - x_0} = f'(x_0)$$

f is 1-time differentiable but $f \notin C^1(\mathbb{R})$.

$$g(x) = \begin{cases} x^3 \cos\frac{1}{x} & , x \neq 0 \\ 0 & , x = 0 \end{cases}$$

$x \neq 0$, $g'(x) = 3x^2 \cos\frac{1}{x} + x \sin\frac{1}{x}$ $\quad x \neq 0$

$$g'(0) = \lim_{x\to 0} \frac{g(x) - g(0)}{x - 0} = \lim_{x \to 0} \frac{x^3 \cos\frac{1}{x} - 0}{x} = 0$$

$$g'(x) = \begin{cases} 3x^2 \cos\frac{1}{x} + x\sin\frac{1}{x} & , x \neq 0 \\ 0 & x = 0 \end{cases}$$

$g \in C^1(\mathbb{R})$

~~$x \neq 0$, $g'(x) = 3x^2\cos\frac{1}{x} + x\sin\frac{1}{x}$, $x \neq 0$~~

$g'(0) = \lim\limits_{x \to 0} g(x)$

Def: we say that f is infinitely differentiable on Ω if $f \in \bigcap\limits_{n \in \mathbb{N}_0} C^n(\Omega)$ and we denote by $f \in C^\infty(\Omega)$.

Remark.

When we say f is n-time continuously differentiable, we mean that $f \in C^n$

theorem: Let $f, g : \Omega \to \mathbb{R}$ be two functions infinitly diff. then:

$$(f(x)g(x))^{(n)} = \sum_{k=0}^{n} \binom{n}{k} f^{(k)}(x)\, g^{(n-k)}(x)$$

Example:

$$(\sin e^x)^{(10)} = \sum_{k=0}^{10} C_n^k (\sin x)^{(k)} \cdot e^{x(10-k)}$$

$$= e^x \sum_{k=0}^{10} C_{10}^k \sin\left(x + k\frac{\pi}{2}\right)$$

Convex functions:

$f, [a, b] \to \mathbb{R}$

we say that f is convex on $[a, b]$, if

$\forall x, y \in [a, b]$: [illegible]

paramezation for $[x, y]$

$(1-\lambda) f(y)$

[illegible]

not convex = [illegible]

theorem: Let f be a convex function on $[a, b]$; then, $\forall (x_i)_{i=1..n} \in [a, b]$, $\forall \lambda_i \in [0, 1]$: $\lambda_1 + \dots + \lambda_n = 1$ we have: (generalization)

$$f\left(\sum_{k=1}^{n} \lambda_k x_k\right) \leq \sum_{k=1}^{n} \lambda_k f(x_k)$$

proof: $n = 1$ (obvious)

$n = 2$ true (definition)

means: $f\left(\sum_{k=1}^{n} \lambda_k x_k\right) \leq \sum_{k=1}^{n} \lambda_k f(x_k)$

and let us prove for $(n+1)$

$$f\left(\sum_{k=1}^{n} \lambda_k x_k + \lambda_{n+1} x_{n+1}\right)$$

$$= f\left(\sum_{k=1}^{n} \lambda_k \cdot \sum_{k=1}^{n} \frac{\lambda_k}{\sum_{k=1}^{n} \lambda_k} x_k + 1 \lambda_{n+1} x_{n+1}\right)$$

③ $\forall x,y,z \in [a,b]: \; x<y<z$

$$\frac{f(y)-f(x)}{y-x} \leq \frac{f(z)-f(y)}{z-y}$$

④ $\forall x,y,z \in [a,b]: \; x<y<z$

$$\frac{f(z)-f(x)}{z-x} \leq \frac{f(z)-f(y)}{z-y}$$

<u>proof:</u> we prove $(1) \Leftrightarrow (2)$

$1) \Rightarrow 2)$

we suppose f is convex.

$$y = \frac{z-y}{z-x}\cdot x + \frac{y-x}{z-x}\cdot z$$

$$\boxed{\frac{z-y}{z-x} + \frac{y-x}{z-x} = 1}$$

$$\Rightarrow f(y) = f\left(\frac{z-y}{z-x}\cdot x + \frac{y-x}{z-x}\cdot z\right)$$

$$\leq \frac{z-y}{z-x}\cdot f(x) + \frac{y-x}{z-x} f(z)$$

by adding $-f(x)$ to both sides of the inequality

$$f(y)-f(x) \leq \frac{y-x}{z-x}\left(f(z)-f(x)\right)$$

$$\rightarrow \frac{f(y)-f(x)}{y-x} \leq \frac{f(z)-f(x)}{z-x} \quad ✓$$

$(2) \Rightarrow (1)$ $\quad \forall x,y,z \in [a,b] \; (x<y<z): \; \dfrac{f(y)-f(x)}{y-x} \leq \dfrac{f(z)-f(x)}{z-x}$

let $x_1 < x_2$ $\quad (x_1, x_2 \in [a,b])$

$x = x_1$, $\quad y = \lambda x_1 + (1-\lambda)x_2$, $\quad z = x_2$

one has

$$\frac{f(\lambda x_1 + (1-\lambda)x_2) - f(x_1)}{\lambda x_1 - x_1 + (1-\lambda)x_2} \leq \frac{f(x_2)-f(x_1)}{x_2-x_1}$$

proof: $n=1$ (obvious)

$n=2$ true (definition) (bc it's convex)

means: $f\left(\sum_{k=1}^{n}\lambda_k x_k\right) \le \sum_{k=1}^{n}\lambda_k f(x_k)$

and let us prove for $(n+1)$ ($p(n)$ is assumed true)

$$f\left(\sum_{k=1}^{n}\lambda_k x_k + \lambda_{n+1} x_{n+1}\right) \quad \text{the sum} = 1$$

$$= f\left(\underbrace{\sum_{k=1}^{n}\lambda_k}_{\alpha} \cdot \underbrace{\sum_{k=1}^{n}\frac{\lambda_k}{\sum_{k=1}^{n}\lambda_k}x_k}_{x} + \underbrace{1\lambda_{n+1}}_{1-\alpha}\underbrace{x_{n+1}}_{y}\right) \le \sum_{k=1}^{(n-2)n}\lambda_k f\left(\sum\frac{\lambda_k}{\sum\lambda_k}x_k\right) + \lambda_{n+1}f(x_n)$$

(constant) Real number

we use the assumption and the fact sum=1

$$\left(\sum_{k=1}^{n}\frac{\lambda_k}{\sum\lambda_k} = 1\right)$$

$$\overset{p(n)}{\le} \sum_{k=1}^{n}\lambda_k\left(\sum_{k=1}^{n}\frac{\lambda_k}{\sum_{k=1}^{n}\lambda_k}f(x_k)\right) + \lambda_{n+1}f(x+1)$$

$$\le \sum_{k=1}^{n}\lambda_k f(x_k) + \lambda_{n+1} f(x_{n+1})$$

theorem:

Let $f:[a,b] \to \mathbb{R}$

the following statements are equivalent:

1. f is convex in $[a,b]$
2. $\forall x,y,z \in [a,b]: x<y<z$

$$\frac{f(y)-f(x)}{y-x} \le \frac{f(z)-f(x)}{z-x}$$

$$\Rightarrow \frac{f(\lambda x_1 + (1-\lambda)x_2) - f(x_1)}{(1-\lambda)} \leq f(x_2) - f(x_1)$$

$$\Rightarrow f(\lambda x_1 + (1-\lambda)x_2) \leq (1-\lambda)f(x_2) - (1-\lambda)f(x_1) + f(x_1)$$

$$\Rightarrow f(\lambda x_1 + (1-\lambda)x_2) \leq \lambda f(x_1) + (1-\lambda)f(x_2)$$

~~coro~~ corollary:

Let $f : [a, b] \to \mathbb{R}$ be function : the following are equivalent:

1) f is convex

2) $\forall a \in \mathbb{R} : \varphi_a : \mathbb{R} \setminus \{a\} \to \mathbb{R}$

$$x \mapsto \varphi_a(x) = \frac{f(x) - f(a)}{x - a}$$

is nondecreasing

Remark: Let $f : [a, b] \to \mathbb{R}$ such $f^{(2)}$ exists. then:

f convex $\Leftrightarrow f'' > 0$

f concave $\Leftrightarrow f'' < 0$

$a > 0,\ b > 0,\ p > 0,\ q > 0,\ \frac{1}{p} + \frac{1}{q} = 1$

$$ab \leq \frac{a^p}{p} + \frac{b^q}{q}$$

to prove: let $f(x) = \ln x$, $f^{(2)}(x) = -\frac{1}{x^2} < 0$ $]0, +\infty[$

$$\ln\left(\frac{1}{p}a^p + \frac{1}{q}b^q\right) \geq \frac{1}{p}\ln a^p + \frac{1}{q}\ln b^q$$

$$\geq \ln a + \ln b$$

$$\geq \ln ab$$

$$ab \leq \frac{a^p}{p} + \frac{b^q}{q}$$